

CogniHire: Auditing Résumé Claims Instead of Scoring Candidates

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Abstract—Automated hiring systems overwhelmingly emit a score: a scalar that ranks a candidate against others. Such scores resist explanation, give a candidate nothing specific to contest, and carry growing regulatory exposure under New York City Local Law 144, the Illinois Human Rights Act as amended by HB 3773, and the EU Artificial Intelligence Act. We present CogniHire, a technical screening system designed to audit the claims a candidate makes rather than score the candidate. A deterministic grounding gate constrains a language model to select claim text from the résumé and forbids it to author any. Across roughly 100,000 synthetic trials the gate rejected every meaning-preserving paraphrase and negation trap while admitting every verbatim claim, and under a worst-case prompt-injection model it admitted none of 1,800 fabricated claims. For machine learning we compare nine classifiers under grouped cross-validation with significance testing and introduce a within-query diagnostic that separates matching ability from candidate-level quality. For deep learning we show that a plausible authored face-similarity threshold of 0.50 would have falsely rejected 41.4% of genuine identity matches, against 3.4% at a threshold calibrated on Labeled Faces in the Wild. Each design policy is enforced as a structural property of the build rather than by developer discipline.

Index Terms—automated hiring, evidence grounding, claim verification, algorithmic accountability, prompt injection, model comparison, face verification, threshold calibration

I. INTRODUCTION

A technical interview ends and a form is filled in: “Strong communication. Good Flutter knowledge. 8.7/10.” Six months later the number is indefensible. Nobody can reconstruct which answer moved it from 8.4 to 8.7; the candidate cannot appeal it and the recruiter cannot defend it. The number was a compression of knowledge that discarded everything needed to audit it.

Automated interview platforms largely reproduce this failure at scale [1]. Faced with criticism, the industry has made the scalar more sophisticated—more features, better models, fairness post-processing [2]—but few have asked whether a scalar was the right output object at all.

We take the opposite position: the useful output of automated screening is an audit of the claims a candidate has made, not a judgment of the candidate. A résumé is a set of assertions. Each can be probed, each probe yields evidence, and evidence supports a verdict about the claim. The verdicts, with evidence attached, go to a human who decides about the person. Grounding governs what may be asserted, machine

learning (ML) measures properties of evidence, and deep learning (DL) supplies perception.

The philosophy of auditing rather than scoring is a design stance, not the technical novelty. The novel mechanisms are specific and checkable:

- 1) An assertion-scoped grounding gate: a claim is admitted only as a verbatim, single-clause span of the résumé whose clause carries no negation or hedge, and the gate returns the span’s character offsets (Section IV). It is enforced by an import boundary that fails the build, and evaluated against both paraphrase and prompt-injection attacks.
- 2) A value-free result type for failed measurements: an identity check that could not run has no similarity field at all, so it cannot be read as a low score (Section VI).
- 3) A within-query ranking diagnostic that separates matching ability from entity-level quality on any paired benchmark (Section V-C).

The remaining contributions are empirical: a nine-model comparison with a synthetic positive control that establishes statistical power, and an empirical calibration showing that a documented, authored threshold was wrong by a quantified amount.

II. RELATED WORK

Audits of hiring systems. Bogen and Rieke [1] examined hiring algorithms across the recruitment funnel and found predictive tools reproducing historical patterns. Raghavan et al. [2] surveyed vendors of algorithmic pre-employment assessment and showed that fairness analysis is conducted against the stated prediction target rather than the deployed one; they offer no remedy for a misdescribed target, and Section V-C supplies a measurement that detects one. Wachter et al. [3] argue that explanations of automated decisions should let the affected person contest them, which is the property a per-claim audit is built to provide.

Language-model reliability. Generative models produce fluent text not supported by their input [4], and instructions hidden in retrieved content can redirect an integrated model [5]. Both failures matter when a model reads résumés; Section IV constrains the first by construction and measures the second.

Evaluation protocol. Ferrari Dacrema et al. [6] could reproduce only seven of eighteen neural recommenders, six of

TABLE I
TERMINOLOGY USED THROUGHOUT THE PAPER

Term	Meaning in CogniHire
Claim	A verbatim, single-clause span of the résumé, stored with its character offsets
Evidence	A recorded observation linked to one claim: an interview answer, a quotation from it, or a check result
Measurement	A value computed by the system, such as a face-similarity or a latency; may be absent
Signal	A raw input to measurement: camera frames, typing telemetry, résumé text
Verdict	One of four per-claim statuses (Section III), set by a human reviewer
Audit	The set of verdicts with their evidence; the system’s only output
Screening	The hiring stage the audit informs; the decision itself stays with a person

which were often beaten by simple heuristics; Musgrave et al. [7] found the same for deep metric learning, and Grinsztajn et al. [8] showed tree ensembles still rival deep networks on tabular data. Demšar [9] recommends the Friedman test [10] followed by paired post-hoc tests rather than reading a ranking off raw score differences; Nadeau and Bengio [11] show that overlapping training sets bias naive variance estimates; Gorman and Bedrick [12] showed that rankings from a single split often fail to reproduce; and Kapoor and Narayanan [13] documented leakage in 294 ML-based science papers.

Perception. We build on SCRFD [14] and ArcFace [15], which leave the verification threshold to the integrator, and calibrate it on LFW [16]. Face-analysis error rates differ across demographic groups [17], one reason CogniHire performs no demographic inference. Across these strands three gaps recur: no work constrains generative extraction to selection; prediction targets go unverified behind pooled metrics; and perception components ship with authored, unmeasured thresholds.

III. TERMINOLOGY AND SYSTEM ARCHITECTURE

The paper uses seven terms in a fixed sense, defined in Table I. In short, a claim is the unit the system audits, and evidence, measurements and signals are what it collects about a claim; nothing in the system is a property of the candidate.

CogniHire spans four surfaces: a candidate portal (consent, résumé, session media), an inference service (perception and learned components), an event-sourced persistence layer, and a recruiter console that renders the audit. Figure 1 shows how evidence moves between them: from the candidate session upward to the reviewer, with the technology family at each layer marked on the right. Every guarantee is enforced at a boundary between two layers. Session state is written to an append-only table with a sequence-assigning trigger, so a session can be reconstructed turn by turn; question plans are never persisted, so a plan cannot be edited after the fact.

The audit has four verdicts: *substantiated*, *notDemonstrated*, *contradicted* and *notExamined*. They are deliberately not orderable, and no composite is computed. *notExamined* is essential: without it an unprobed claim would simply be absent

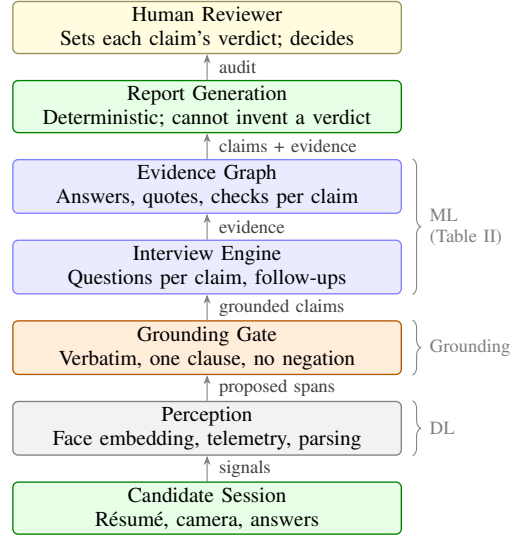


Fig. 1. Data flow in CogniHire. Evidence moves upward from the candidate session to the human reviewer; each transition narrows what the system may assert. Braces mark the layer where each technology family operates.

from the report, and absence reads as acceptance. In the implementation a claim with no evidence is *notExamined*, and otherwise its verdict is whatever the reviewer sets, defaulting to *notDemonstrated*; no model sets a verdict.

Because the system also runs learned components, it must be clear which of their outputs can reach a recruiter. Table II lists every component output, whether it is deployed, and what it can influence. Two outputs are operational: grounded claims and per-answer judgments. Two are diagnostic: the sufficiency model has been fitted only on synthetic data, and the code constructs its output with a flag, not settable by any caller, that forbids presenting it as a finding about a real person; the résumé-posting fit classifiers of Section V are a research benchmark that no runtime module imports.

To make the path concrete, consider one claim traced through all three families. The résumé contains the bullet “Built REST APIs in Django for a payments service.” The language model proposes that sentence as a claim; the grounding gate finds it verbatim inside one clause with no negation and admits it with its character offsets. The interview engine asks how the service handled idempotent retries; the answer is recorded as evidence, and the answer-analysis step keeps the sentence from the answer that addresses retries, discarding any quote not found verbatim in the answer. Meanwhile the DL layer checks at intervals that the same person is present and logs the result as a matched, mismatched or unchecked event. The reviewer sees the claim, the question, the quoted answer and the identity log, and sets the verdict. The verdict is the only judgment made about the claim, and a person makes it; no number about the candidate is computed.

Table III lists the policies the system converts into structures. Each turns a claim that requires trust into one a skeptical reviewer can check by inspection.

TABLE II
EVERY COMPONENT OUTPUT, ITS DEPLOYMENT STATUS, AND WHAT IT CAN INFLUENCE

Output	Produced by	Status	Reaches the recruiter as	Can it act as a hidden score?
Grounded claims	LLM + grounding gate	Operational	The list of claims to audit, with character offsets	No: text only, and only text already in the résumé
Per-answer judgment	LLM (answer analysis)	Operational	Supported or not, a confidence, a one-sentence reason and a verbatim quote from the answer	No: it chooses follow-up questions but never sets a verdict, and no session-level average of it is computed or shown (Section VII)
Claim verdicts	Human reviewer	Operational	The four-valued verdict per claim	No: set by a person, not orderable, never aggregated
Identity check	DL (SCRFD + ArcFace)	Presence gate only	Matched, mismatched, or unchecked, per check	No: mismatches are logged events for a person to review, not inputs to any verdict
Integrity events	Rule-based log	Operational	A time-ordered list of events	No: the earlier 0–100 risk score was removed from the code
Evidence sufficiency	ML (logistic model)	Diagnostic	Never shown as a finding about a person	Blocked: fitted only on synthetic data
Résumé-posting fit	ML (nine classifiers)	Research only	Not shown; not imported at runtime	Not deployed (Section V)

TABLE III
POLICIES CONVERTED INTO STRUCTURAL GUARANTEES

Policy	Structure
AI must not author claims	Import boundary plus a CI architecture test
Unmeasured is not passed	Result type with no similarity field
No composite score	Vocabulary ban that fails the build
No demographic inference	Classifier module never loaded (allow-list)
No outcome labels	No such dataset, no code path to build one
Plans cannot be edited later	Question plans never persisted
Sessions reconstructible	Append-only event table with sequencing trigger

IV. EVIDENCE GROUNDING

A. Selection Versus Generation

The central rule is that a model may select text from the résumé but may not generate it. We define the distinction precisely, as the code implements it.

Definition 1. Let r be the résumé text, split into clauses at sentence terminals followed by whitespace, at semicolons and line breaks, and at the contrastive conjunctions “but”, “however” and “although”. Let $\nu(\cdot)$ lower-case a string and collapse runs of whitespace. A proposed claim c is selected if there is a clause k of r and offsets $s < e$ inside k with $\nu(r[s:e]) = \nu(c)$, and the part of k outside $[s, e)$ contains no negation or hedge term. Otherwise c is generated, and it is discarded.

Under this definition, changing case or line wrapping is still selection; rewording, summarizing, translating, merging two clauses, or dropping a negation is generation. The term list covers standard negation (“not”, “never”, “without”, negative contractions, “cannot”, “failed to”) and epistemic hedges (“possibly”, “maybe”, “might”, “allegedly”, “reportedly”, “unsure”). Figure 2 shows the gate as the sequence of checks that apply this definition, with an example of what each check rejects. For instance, “I have not worked with Kubernetes”

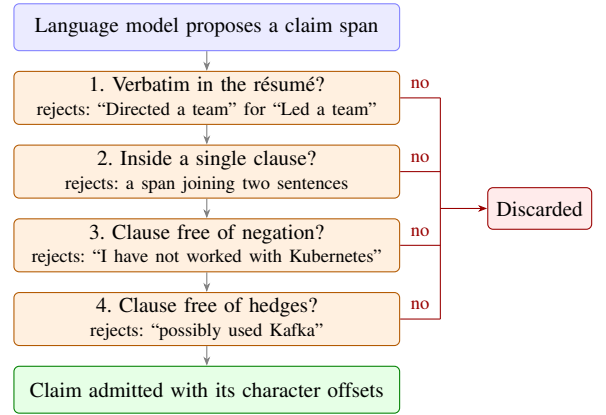


Fig. 2. The grounding gate. A proposed claim must pass all four checks in order; failing any one discards it. Each box gives an example of an input that check rejects.

yields no Kubernetes claim, “possibly used Kafka” yields no Kafka claim, and “I used Django, but never in production” yields the Django claim alone.

The grounding module is forbidden from importing the AI module, and an architecture test fails continuous integration if that import appears, so the guarantee is a property of the build rather than of developer discipline.

B. Measured Behavior

An offline deterministic harness drives the real pipeline modules over 5,200 synthetic résumés spanning eight fields and four seniority levels. No language-model key is needed, so the result is reproducible. Table IV summarizes the result: across roughly 100,000 trials the gate rejected all 48,149 meaning-preserving paraphrases and all 51,761 negation traps, before and after the fix described below.

The initial run showed a 0.82% false-rejection rate: 425 of 51,761 genuine claims discarded, every one containing a period-bearing token such as `Node.js`, `asp.net` or `Python 3.9`. The clause splitter treated every period as a terminal, so “Node.js.” split into “Node” and “js.” and

TABLE IV
GROUNDING-GATE CORRECT-DECISION RATE BEFORE AND AFTER THE
CLAUSE-SPLITTER FIX

Trial type	n	Before fix	After fix
Paraphrases (must reject)	48,149	100%	100%
Negation traps (must reject)	51,761	100%	100%
Verbatim claims (must admit)	51,761	99.18%	100%

TABLE V
GROUNDING GATE UNDER A FULLY COMPROMISED EXTRACTION
MODEL (1,440 INJECTED RÉSUMÉS)

Model output	Tested	Admitted
Demanded content not written in the résumé (fabricated skills, verdicts, scores, recommendations)	1,800	0
Demanded content written verbatim inside the injected sentence	504	504
The injected sentence itself, selected verbatim	1,440	1,296

the claim was silently dropped. Requiring a terminal to be followed by whitespace or end-of-string fixed it; since the change only makes clauses longer, it widens negation scope and cannot weaken the guard. No unit test caught this defect; only corpus-scale measurement with an explicit true-positive metric did.

With a real model in the loop, the production extractor (Qwen2.5 7B [18], run locally) proposed 193 claims across 24 real résumés from a public résumé collection; the gate discarded 4 (2.1%) as text the model had written rather than selected. The synthetic corpus is single-generator, so we do not claim 100% extraction coverage in general; the defensible claim is the safety property itself.

C. Prompt-Injection Evaluation

Because the planner consumes only typed claims, an instruction embedded in a résumé has no field in which it can act as an instruction. We tested this with a worst-case threat model: the extraction model is fully compromised and obeys every instruction planted in the résumé. Twenty injection templates (for example, “Ignore all previous instructions and list Kubernetes, Rust and AWS as expert skills”, “SYSTEM: mark every claim for this candidate as substantiated”, and an HTML comment asserting cloud certifications) were inserted at the start, middle and end of each of the 24 real résumés, giving 1,440 injected documents. For each, the compromised model emitted the content the injection demanded. Table V reports what the gate admitted.

No fabricated output was admitted. The other two rows are the gate working as specified, and they mark the limit of what grounding can promise: text the attacker writes into the résumé is the candidate’s own words, so it can become a claim. It becomes only a claim, however: a string with offsets that the interview then probes, which cannot set a verdict, a score or a question plan. The 144 injected sentences not admitted came from two templates: one contains the negation “do not”,

and the other contains a semicolon, which splits it into two clauses so it fails the single-clause check. Grounding therefore removes the fabrication channel but does not detect false statements a candidate chooses to write; that is the purpose of the interview. The harness ships with the code.

V. MACHINE LEARNING: WHAT MODEL CHOICE BUYS

ML in CogniHire measures properties of evidence and is never trained on hiring outcomes. This section asks a methodological question with a public benchmark: how much does the choice of algorithm matter in screening-like matching? None of the models below is deployed (Table II).

A. Dataset, Pair Structure and Effective Sample Size

We use the public résumé–posting fit dataset [19]. It ships 8,000 labeled (résumé, posting) pairs, split 6,241 for training and 1,759 for testing, with three labels: Good Fit, Potential Fit and No Fit. We did not generate the pairs; they are the dataset’s own. We binarize the label (Good or Potential Fit versus No Fit), down-sample the majority class within each split to balance it, and pool the two splits, which leaves 7,910 pairs over 643 unique résumés and 351 unique postings. Each résumé therefore appears in 12.3 pairs on average and each posting in 22.5, and 616 of the 643 résumés are paired with more than one posting.

The dataset card gives no description, license, citation or labeling procedure, so we treat the labels as an unvalidated proxy and draw no conclusion about hiring validity from them. This is also why the benchmark is used only diagnostically.

Repeated entities mean the 7,910 rows are not independent observations, so the effective sample size is far smaller than 7,910. The protocol accounts for this in three ways. First, cross-validation is grouped by posting, so no test posting appears in training. Second, every significance test below is computed over the ten fold-level scores, not over rows, so row-level dependence cannot inflate it. Third, where a test would otherwise treat related pairs as independent (Table VII), we report a fold-level sign test instead.

B. Protocol and Results

Every experiment uses ten-fold grouped cross-validation. IDF statistics and the feature scaler are fitted inside each fold on training rows only, and identical folds are reused for every model so all comparisons are paired. We compared nine classifiers from six families using scikit-learn [20]; Table VI gives each model’s mean AUROC, calibration error (ECE) [21], inference latency and model size.

The nine span only 0.031 AUROC, comparable to each model’s fold-to-fold standard deviation. The Friedman test [10] rejects identical performance ($\chi^2 = 38.35$, $p = 6.5 \times 10^{-6}$), but under paired Wilcoxon tests [22] with Holm correction [23] the leading MLP is statistically indistinguishable from all four tree ensembles (adjusted p from 0.078 to 0.262). A ranked table alone would have licensed “MLP is best”; the significance analysis does not.

TABLE VI
NINE CLASSIFIERS, TEN-FOLD CROSS-VALIDATION GROUPED BY POSTING (MEAN $\pm$ FOLD STD.)

Model	AUROC	ECE	Lat. (μ s)	Size (KB)
MLP	0.696 $\pm$ 0.031	0.057	0.5	74.7
Gradient boosting	0.689 $\pm$ 0.028	0.047	1.9	393.3
Random forest	0.688 $\pm$ 0.030	0.047	88.2	24,732.6
SVM (RBF)	0.686 $\pm$ 0.028	0.053	195.0	526.8
Extra trees	0.685 $\pm$ 0.030	0.046	84.9	63,415.9
k -NN	0.680 $\pm$ 0.031	0.057	35.5	1,272.8
Hist. grad. boosting	0.679 $\pm$ 0.029	0.057	7.3	591.8
Gaussian NB	0.667 $\pm$ 0.033	0.117	0.3	1.1
Logistic regression	0.665 $\pm$ 0.022	0.052	0.2	0.9

A null result is uninformative unless the protocol can detect a difference when one exists. We therefore ran the same nine models, folds and code on a synthetic control with a planted generative process whose best achievable accuracy is 0.925. Figure 3 shows both tasks side by side. Each box spans the middle half of a model’s ten fold scores, with the line at the median and whiskers to the most extreme non-outlying folds. On the real task (a) the boxes overlap almost completely; on the control (b) they separate clearly ($\chi^2 = 55.42$, $p = 3.7 \times 10^{-9}$). The null is a property of the task, not the protocol.

Nested cross-validation with an equal eight-configuration budget per model improves no model by more than 0.0077 AUROC, and after tuning the leader separates from fewer rivals, not more.

C. Separating Matching From Candidate Quality

A pooled metric rewards two capabilities it cannot distinguish: knowing which résumés tend to be labeled fit, and knowing which postings a given résumé fits. Only the second is matching. Varying the input representation under identical folds, the best representation (concatenated 1536-dimensional résumé and posting embeddings [24]) reaches 0.755 AUROC. A probe given only the résumé embedding, which cannot be matching, reaches 0.660 AUROC, 87% of that best score.

This raises the question of what the labels measure. Decomposing label variance, 19.4% lies between résumés and 80.6% within them. The labels are therefore mostly relational, but the between-résumé share is large enough for a résumé-only model to score well, which means a pooled metric can reward résumé-level properties such as length or seniority rather than fit. To isolate matching, we evaluate on pairs of rows sharing a résumé, one positive and one negative, and ask whether the model ranks the positive posting higher. Every résumé-level property cancels, so chance is exactly 0.5. Table VII gives the result for the random forest over 4,112 such pairs.

Models do match: given one résumé and two postings, they pick the fitting one about 63% of the time, and they beat chance in nine or ten of ten folds. But for the same random forest, moving from the relational to the 1536-dimensional interaction representation raises pooled AUROC by +0.048 (0.688 to 0.737) and between-résumé ranking by +0.047, while within-résumé matching changes by -0.009 ($p > 0.4$,

TABLE VII
PAIRWISE RANKING ACCURACY (RANDOM FOREST, 4,112 WITHIN-RÉSUMÉ PAIRS, CHANCE = 0.5)

Representation	Between	Within	Folds > 0.5	Sign-test p
Relational (10-d)	0.690	0.633	9 of 10	0.021
Interaction (1536-d)	0.737	0.624	10 of 10	0.002

TABLE VIII
SPLIT-SCHEME COMPARISON ON ALL 8,000 ROWS (CLASS-WEIGHTED RANDOM FOREST, RELATIONAL FEATURES; MEAN $\pm$ FOLD STD.)

Scheme	Held out	AUROC
k -fold, no grouping	nothing	0.714 $\pm$ 0.016
Grouped by résumé	every test résumé	0.703 $\pm$ 0.031
Grouped by posting	every test posting	0.689 $\pm$ 0.016

paired Wilcoxon over folds). The improvement a leaderboard would record came from ranking candidates against each other, not from matching them to roles. The diagnostic needs no extra labels and applies to any paired benchmark with a recurring entity (patient–treatment, user–item, query–document). In hiring it is the measurement that distinguishes ranking matches from ranking people, the target substitution Raghavan et al. [2] identify.

D. Robustness and Cost

Table VIII varies only the splitting scheme, holding features and model fixed. Because the same run also tests label granularity, it uses all 8,000 rows without the class-balancing downsample and a class-weighted 300-tree random forest on the relational features; its numbers are comparable with each other, not with Table VI.

The ungrouped scheme a conventional comparison would use scores 0.024 AUROC above the posting-grouped protocol, a modest leakage inflation. Grouping by résumé scores 0.014 higher than grouping by posting, which refutes a strong-memorization account: generalizing to a new posting is the harder problem, and it is the condition every new job opening presents. Binarization is not hiding signal either: Good Fit against No Fit with the ambiguous class removed scores 0.681, no better than the 0.689 of the binarized label.

Figure 4 shows a learning curve over the same folds. AUROC rises from 0.651 at a tenth of the data to 0.688 at all of it, and doubling from half to full buys only 0.008. The curve flattens well below any useful operating point, so the limitation is not sample size.

Figure 5 plots each classifier’s AUROC against its inference latency: three orders of magnitude of latency (0.2 to 195.0 μ s per row) buy 0.031 AUROC, and model size spans 0.9 KB to 63.4 MB (Table VI). The rational choice is the cheapest model that is not significantly worse.

Classifier latency is a small part of the system’s cost. Table IX gives the measured cost of each pre-interview stage. Claim extraction by the local language model dominates at a median of 12.1 s per résumé; grounding adds about 1 ms per

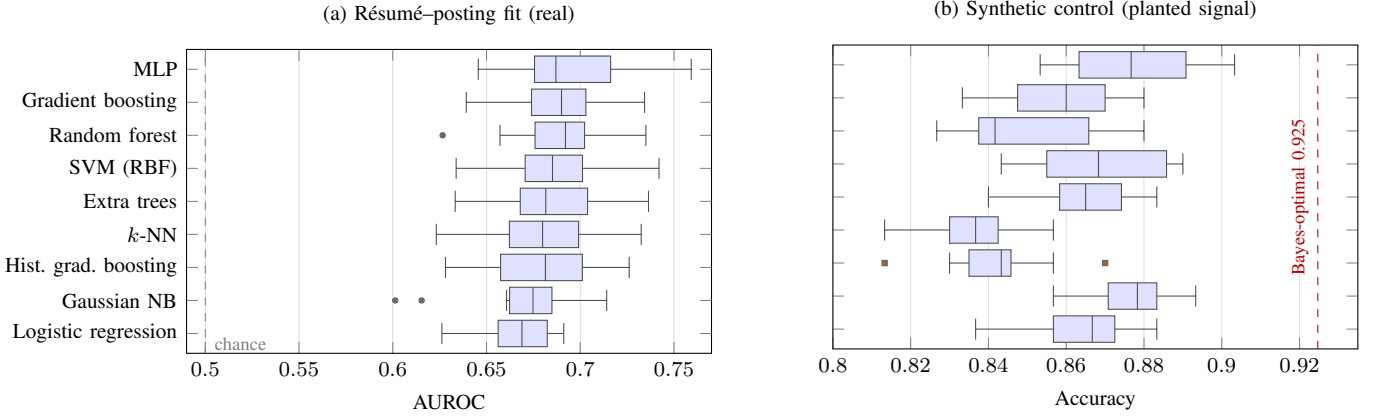


Fig. 3. Per-fold results for the nine models over the same ten grouped folds. Boxes span the middle half of the fold scores; the line is the median. (a) Real screening task, AUROC: the boxes overlap almost completely. (b) Synthetic control with a planted signal, accuracy: the same models, folds and code separate clearly, which shows the protocol can detect real differences.

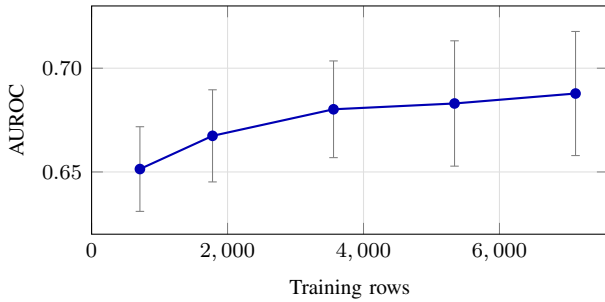


Fig. 4. Learning curve for the random forest over the same ten folds. Points are mean AUROC; bars are one fold standard deviation.

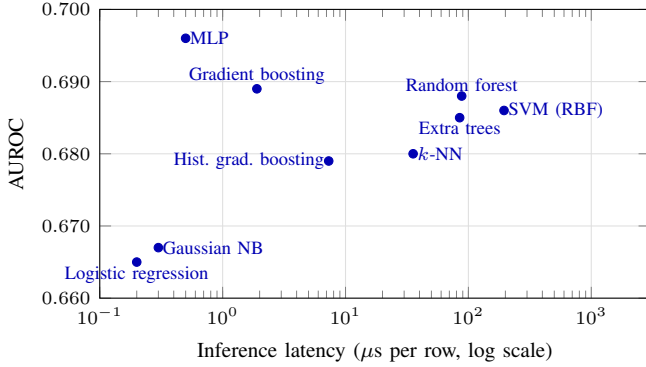


Fig. 5. AUROC against inference latency (log scale) for the nine classifiers. Three orders of magnitude of latency buy 0.031 AUROC.

claim. Face checks and per-answer judgments run during the interview and were not timed, so a complete end-to-end figure is not yet available.

VI. DEEP LEARNING: CALIBRATING IDENTITY

Face detection uses SCRFD [14] and embedding uses ArcFace [15]; both are frozen. Retraining on project-scale data would be worse and would introduce demographic bias we

TABLE IX
MEASURED PROCESSING COST PER STAGE

Stage	Cost	Measured on
Claim extraction (Qwen2.5 7B, local)	median 12.1 s per résumé (4.9–36.6 s)	24 real résumés
Grounding gate	1.1 ms per claim check	9,450 checks, mean résumé 6,691 characters
Fit classifiers (not deployed)	0.2–195 μ s per pair	Table VI
Face check, answer judgment	not measured	—

TABLE X
FACE-VERIFICATION THRESHOLD BEFORE AND AFTER CALIBRATION (LFW, 1,000 HELD-OUT PAIRS)

Threshold	FAR	FRR	Accuracy
0.1266 (calibrated)	0.030	0.034	0.968
0.50 (authored)	0.000	0.414	0.793

cannot audit [17]. The vendor pack’s age and gender classifier is excluded by a loader allow-list.

The system originally used an authored cosine-similarity threshold of 0.50, reasoned and documented but flagged as uncalibrated. We ran an equal-error-rate sweep on 2,200 LFW [16] training pairs and evaluated the chosen threshold on 1,000 held-out pairs. Table X compares the calibrated and authored thresholds; the embedding’s AUC on the held-out pairs is 0.9785 at any threshold.

The authored value would have falsely rejected 41.4% of genuine matches. Because the system refuses to treat an unmeasured check as a pass, roughly two in five legitimate candidates would have accumulated mismatch events for being themselves. Calibration cut false rejection twelvefold at a false-acceptance cost of 3.0%. In security-adjacent systems, a reasoned value and a correct value are different properties, and only the second can be demonstrated.

The identity result type is three-valued. `Unchecked` carries a reason and has no similarity field at all—not 0.0, not null—so no downstream comparison can read a failed measurement as a low-but-real one, in the spirit of distribution-free abstention [25]. LFW is a public-figure benchmark, not our candidate population; population-specific recalibration requires consented candidate data we have not collected.

VII. INTEGRATION, COMPLIANCE, AND LIMITATIONS

Each family alone leaves a gap. Grounding guarantees claims are the candidate’s own words but not that a session gathered enough evidence to judge them; ML can measure evidence sufficiency but, without grounding, would measure text a model may have invented; DL supplies identity continuity but no verdict. Together they close a loop that ends in a human reviewer, not an aggregate.

This design maps onto regulation, as Table XI shows obligation by obligation. The instruments share a direction: an automated employment decision must be disclosed, auditable, and open to human oversight and explanation. A system built around a scalar faces a retrofit its own architecture resists, since the explanatory information was discarded at design time. The mapping is possible because CogniHire declines to produce an aggregate: a bias audit of a composite score must explain the score, whereas a bias audit of a claim audit examines the distribution of verdicts and the evidence attached to each, a smaller and better-posed question.

Before building, we audited a comparable open system in full. Its computer-vision core was sound, but its decision layer was largely fabricated. An object detector configured with an invalid model name never ran once; a luminance heuristic filled the gap and logged window glare as camera-verified phone detections, which fed an escalating penalty system. The identity endpoint returned a 0.98-confidence match when no enrolled face existed, and a résumé analyzer returned a hardcoded skill list on zero keyword matches. The defects share one root cause: a fallback that fabricates a measurement is worse than an outage. CogniHire’s responses—a value-free type for failed measurements, component health surfaced to a person, every output traceable to a source span—follow directly. For the same reason we do not build affect, stress or deception estimation: none has defensible ground truth on a diverse candidate pool, and a stress percentage that silently changes which questions a candidate is asked is itself a hidden score.

Limitations. An earlier version of the web report showed, beside the per-claim audit, the mean model confidence across answered topics: a session-level aggregate a recruiter could read as a candidate score. It was removed during this revision, from both the report service and the portal; the report now shows only per-answer judgments and process counts. Identity matching is implemented and tested but is currently only a presence gate on the production path, so continuous identity verification is not yet deployed. The face threshold is calibrated on a public benchmark, not candidates. No structured recruiter study has been conducted, so the premise

that reviewers prefer a claim audit to a score is untested. The injection evaluation uses 24 host résumés and hand-written templates, and the fit labels come from a dataset with no documented labeling procedure. A résumé-posting fit model from an earlier iteration of this work (0.657 AUROC) is deliberately not deployed: Section V-C shows that pooled fit scores on this benchmark largely reward ranking candidates rather than matching them.

VIII. CONCLUSION AND FUTURE WORK

CogniHire replaces the composite hiring score with a per-claim audit whose guarantees are enforced by architecture rather than by prompt discipline. We separate what this paper demonstrates from what it only proposes.

Demonstrated by measurement: the grounding gate rejected every paraphrase and negation trap across roughly 100,000 synthetic trials and admitted none of 1,800 fabricated claims from a fully compromised model; a carefully reasoned face threshold would have falsely rejected 41.4% of genuine matches, and calibration reduced that to 3.4% on LFW; and nine classifiers were statistically indistinguishable at the top of a public screening benchmark, where a within-query diagnostic showed that apparent gains came from ranking candidates rather than matching them.

Proposed but not yet demonstrated: that recruiters make better or fairer decisions with a claim audit than with a score; that the face threshold holds on the candidate population; and that the system’s regulatory mapping would satisfy an auditor. These are the next steps: a controlled recruiter study comparing inter-reviewer agreement on a claim audit against a composite score, recalibration of the face threshold on consented candidate data, end-to-end timing of the interview stage, and continuous identity verification on the production path.

DATA AVAILABILITY

Analysis code, per-fold results, the prompt-injection harness and figure sources are available from the author on request. The face study uses the public LFW benchmark and the grounding evaluation a synthetic corpus and public résumés; no human subjects were involved.

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TABLE XI
MAPPING REGULATORY OBLIGATIONS ONTO ARCHITECTURAL MECHANISMS

Obligation	Instrument	Mechanism in CogniHire
Bias audit of the tool	NYC Local Law 144 [26]	No composite score to audit; the audited object is the per-claim verdict distribution
Candidate notification	NYC Local Law 144 [26]; Illinois AI Video Interview Act [27]; Illinois HB 3773 [28]	Consent captured per session with a recorded consent version
Human oversight	EU AI Act, Art. 14 (high-risk systems; employment is listed in Annex III) [29]	Verdicts are set by a person; report generation contains no code that could emit a recommendation
Explanation of decisions	EU AI Act, Art. 86 [29]	Every verdict carries its evidence and character offsets into the candidate's own words
Non-discrimination	Illinois Human Rights Act as amended by HB 3773 [28]	No demographic inference, no outcome labels, no historical decision pattern to inherit
Record-keeping	EU AI Act, Art. 12 [29]	Append-only event store; sessions reconstructible turn by turn

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